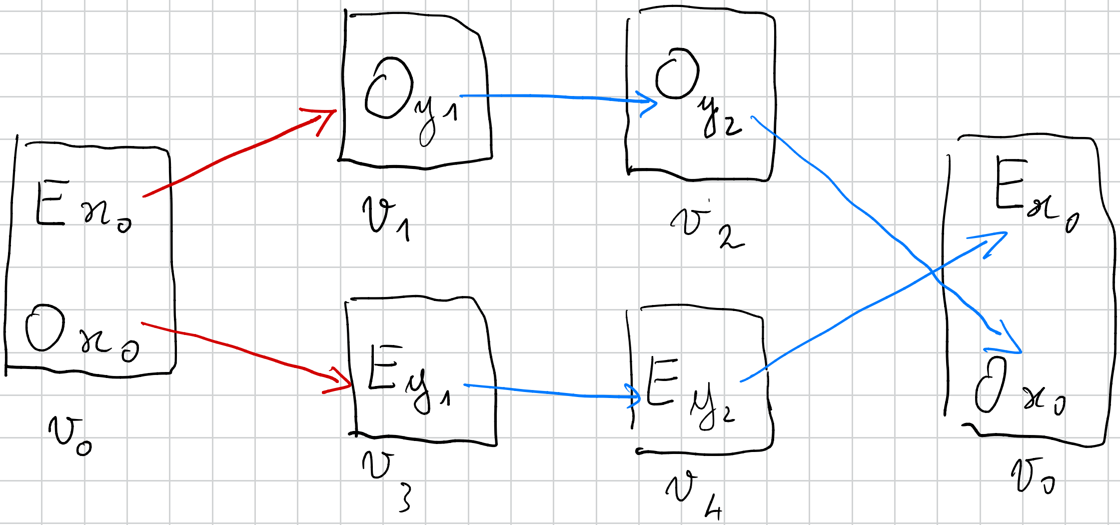
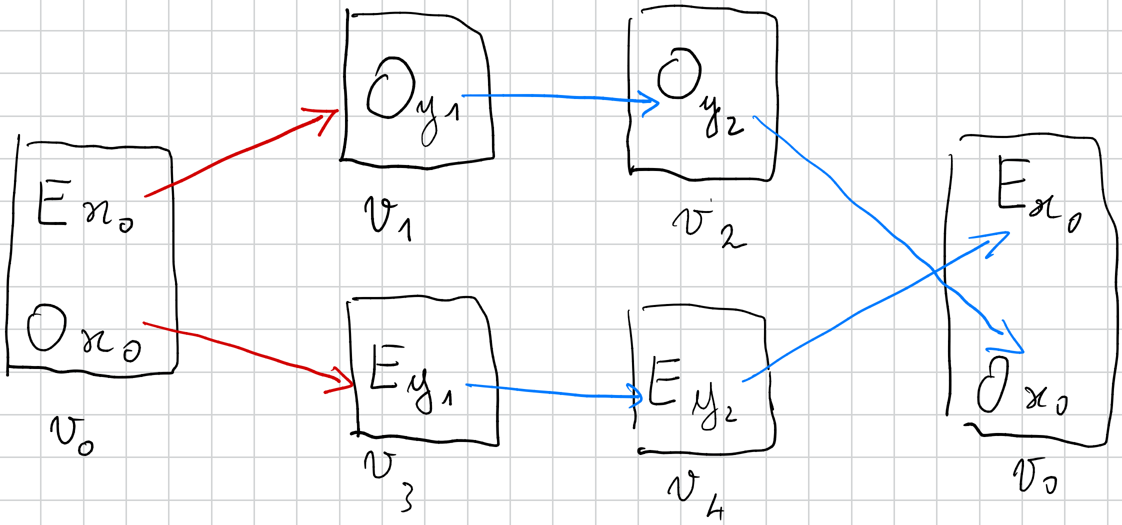
$$
\begin{prooftree}
\AxiomC{}
\RightLabel{(NR_1)}
\UnaryInfC{$\vdash N0$}
\AxiomC{$Ox_0 \vdash Nx\ (\dagger)$}
\RightLabel{(Subst)}
\UnaryInfC{$Oy_2 \vdash Ny$}
\RightLabel{(NR_2)}
\UnaryInfC{$Oy_1 \vdash Nsy$}
\RightLabel{(Case E)}
\BinaryInfC{$Ex_0 \vdash Nx\ (*)$}
\AxiomC{$(*)\ Ex_0 \vdash Nx$}
\RightLabel{(Subst)}
\UnaryInfC{$Ey_2 \vdash Ny$}
\RightLabel{(NR_2)}
\UnaryInfC{$Ey_1 \vdash Nsy$}
\RightLabel{(Case O)}
\UnaryInfC{$(\dagger)\ Ox_0 \vdash Nx$}
\RightLabel{$(\vee L)$}
\BinaryInfC{$Ex \vee Ox \vdash Nx$}
\end{prooftree}
$$

$$X = \{ Ex_0,\ Ey_1,\ Ey_2,\ Ox_0,\ Oy_1,\ Oy_2 \}$$



where Ey_1 is short for $(Ey, 1)$ etc.



$$
\begin{tikzcd}
 & v_1 \arrow[r, "(S_1, P_1)"] & v_2 \arrow[dr, "(S_2, P_2)"] & \\
v_0 \arrow[ur, "(S_0, P_0)"] \arrow[dr, "(S_0', P_0')"] & & & v_0 \\
 & v_3 \arrow[r, "(S_1', P_1')"] & v_4 \arrow[ur, "(S_2', P_2')"] &
\end{tikzcd}
$$

$S_0 = \phi \qquad P_0 = \{(E_{x_0}, O_{y_1})\}$

$S_1 = \{(O_{y_1}, O_{y_2})\} \qquad P_1 = \phi$

$S_2 = \{(O_{y_2}, O_{x_0})\} \qquad P_2 = \phi$

$S_0' = \phi \,, \qquad P_0' = \{(O_{x_0}, E_{y_1})\}$

$S_1' = \{(E_{y_1}, E_{y_2}) \qquad P_1' = \phi$

$S_2' = \{(E_{y_2}, E_{x_0})\} \qquad P_2' = \phi$

Consider the paths:

$$\pi_1 : v_0 \xrightarrow{(S_0, P_0)} v_1 \xrightarrow{(S_1, P_1)} v_2 \xrightarrow{(S_2, P_2)} v_0$$

$$\pi_2 : v_0 \xrightarrow{(S_0', P_0')} v_3 \xrightarrow{(S_1', P_1')} v_4 \xrightarrow{(S_2', P_2')} v_0$$

then there are the infinite paths:

$\pi_1 \pi_1 \cdots \qquad \pi_2 \pi_2 \cdots \qquad$ and

$\pi_1 \pi_2 \pi_1 \cdots \qquad \pi_2 \pi_1 \pi_2 \cdots \qquad$ etc.

that can be described with the generalized regular expression

$$Paths = (\pi_1 + \pi_2)^\omega$$

Let $\pi = \pi_1 \pi_2 \pi_1 \pi_2$ one of these paths; then the trace

$$\tau = Ex_0, Oy_1, Oy_2, Ox_0, Ey_1, Ey_2, Ex_0, \cdots$$

is compatible with π, $\tau \models \pi$, and it is infinitly progressing since

$$Ex_0 \; P_0 \; Oy_1$$

<u>Note</u> the trace τ is compatible only with the traces

$\pi_1 \pi_2 \pi_1 \pi_2 \cdots$ and $\pi_2 \pi_1 \pi_2 \cdots$

the infinite path $\pi_1 \pi_1 \cdots$ has no infinite trace which is compatible with it !!! Similar remark holds for $\pi_2 \pi_2 \cdots$.

If π is a path of $\mathcal{G}$, then define

$$\mathrm{Trace}^{\omega}(\pi) = \{ \tau : \omega \to X \mid \tau \models \pi \}$$

<u>Modified GTC</u>

For all infinite path π, if $\mathrm{Trace}^{\omega}(\pi) \neq \phi$ then there is a trace $\tau \models \pi$ such that τ is infinitely progressing.